

Mathematics for the Slightly Rusty but Still Ambitious

A Practical Review for Interviews, Work, and Future Moments of Panic

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Chapter 1

Multivariable Calculus

1.1 Limits, Continuity, and Derivatives

Definition 1.1.1: Function

A function $f : D \rightarrow E$ is a rule that assigns to each element $x \in D$ exactly one element $f(x) \in E$. The set D is the *domain*, and the set of all actual output values is the *range*.

Increasing and Decreasing Functions Let f be defined on an interval I :

- Whenever $x_1 < x_2$ in I , f is *increasing* on I if $f(x_1) < f(x_2)$, is *nondecreasing* on I if $f(x_1) \leq f(x_2)$.
- Whenever $x_1 < x_2$ in I , f is *decreasing* on I if $f(x_1) > f(x_2)$, is *nonincreasing* on I if $f(x_1) \geq f(x_2)$.

Informal Definition of Limit The statement $\lim_{x \rightarrow a} f(x) = L$ means that $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a , from both the left and the right, but with $x \neq a$.

Right- and Left-Hand Limits $\lim_{x \rightarrow a^+} f(x) = L$ means that $f(x)$ can be made arbitrarily close to L by taking x sufficiently close to a with $x > a$. Similarly, $\lim_{x \rightarrow a^-} f(x) = L$ means that x approaches a from the left, so $x < a$.

Two-Sided Limit The two-sided limit exists and equals L if and only if:

$$\lim_{x \rightarrow a} f(x) = L \iff \lim_{x \rightarrow a^-} f(x) = L \text{ and } \lim_{x \rightarrow a^+} f(x) = L$$

Vertical Asymptote The line $x = a$ is a *vertical asymptote* of the graph of f if at least one of the following is true:

$$\lim_{x \rightarrow a^-} f(x) = \pm\infty, \quad \lim_{x \rightarrow a^+} f(x) = \pm\infty$$

Theorem 1.1.2: Squeeze Theorem

Suppose $g(x) \leq f(x) \leq h(x)$ for all x near a . If $\lim_{x \rightarrow a} g(x) = L$ and $\lim_{x \rightarrow a} h(x) = L$, then $\lim_{x \rightarrow a} f(x) = L$.

Definition 1.1.3: Formal Definition of Limit

Let f be defined near a , except possibly at a . We say $\lim_{x \rightarrow a} f(x) = L$ if for every $\varepsilon > 0$, there exists $\delta > 0$ such that whenever $0 < |x - a| < \delta$, we have $|f(x) - L| < \varepsilon$.

Example 1.1.1

Proof that $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2} = 4$. Let $\varepsilon > 0$ be given. Choose $\delta = \varepsilon$. If $0 < |x - 2| < \delta$, then $x \neq 2$, so $\left| \frac{x^2-4}{x-2} - 4 \right| = \left| \frac{(x-2)(x+2)}{x-2} - 4 \right| = |x + 2 - 4| = |x - 2| < \delta = \varepsilon$. Thus, by the ε - δ definition, $\lim_{x \rightarrow 2} \frac{x^2-4}{x-2} = 4$.

Continuity at a Point A function $f : D \rightarrow \mathbb{R}$ is *continuous at* $a \in D$ if $\lim_{x \rightarrow a} f(x) = f(a)$.

Continuity on an Interval A function is *continuous on an interval* if it is continuous at every point of that interval. At endpoints, continuity is understood using the appropriate one-sided limit.

Types of Discontinuity. Typical discontinuities include:

- *removable discontinuity*: the limit exists but $f(a)$ is missing or has the wrong value;
- *jump discontinuity*: the one-sided limits exist but are not equal;
- *infinite discontinuity*: the function tends to $\pm\infty$;
- *oscillatory discontinuity*: the function oscillates and has no limiting value.

Theorem 1.1.4: Intermediate Value Theorem

Suppose f is continuous on the closed interval $[a, b]$. If N is any number between $f(a)$ and $f(b)$, then there exists at least one $c \in [a, b]$ such that $f(c) = N$.

Limit at Infinity $\lim_{x \rightarrow +\infty} f(x) = L$ means that $\forall \varepsilon > 0$, there exists $N > 0$ such that if $x > N$, then $|f(x) - L| < \varepsilon$. Similarly, $\lim_{x \rightarrow -\infty} f(x) = L$ means that $\forall \varepsilon > 0$, there exists $N > 0$ such that if $x < -N$, then $|f(x) - L| < \varepsilon$.

Horizontal Asymptote The line $y = L$ is a *horizontal asymptote* if

$$\lim_{x \rightarrow +\infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = L$$

Tangent Line If the limit $m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$ exists, then the tangent line to the graph of f at $(a, f(a))$ is

$$y - f(a) = m(x - a)$$

Definition 1.1.5: Derivative at a Point

The derivative of f at a is

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a} = \lim_{h \rightarrow 0} \frac{f(a + h) - f(a)}{h}$$

The derivative is the slope of the tangent line and the instantaneous rate of change of f at a .

Derivative Function The derivative of f is the function defined at all x where this limit exists:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

Differentiability Implies Continuity diffimpliescont If f is differentiable at a , then f is continuous at a .

The converse is false. A function can be continuous but not differentiable. $f(x) = |x|$.

Basic Differentiation Rules Let $c \in \mathbb{R}$ and suppose f, g are differentiable. Then

$$\frac{d}{dx}[c] = 0 \quad \frac{d}{dx}[x^n] = nx^{n-1} \quad \frac{d}{dx}[cf(x)] = cf'(x) \quad \frac{d}{dx}[f(x) \pm g(x)] = f'(x) \pm g'(x)$$

Product Rule and Quotient Rule productrule If f and g are differentiable, then fg is differentiable and

$$\frac{d}{dx}[f(x)g(x)] = f'(x)g(x) + f(x)g'(x) \quad \frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Derivative Formulas for Trigonometric Functions

$$\begin{aligned} \frac{d}{dx} \sin x &= \cos x & \frac{d}{dx} \cos x &= -\sin x & \frac{d}{dx} \tan x &= \sec^2 x \\ \frac{d}{dx} \cot x &= -\csc^2 x & \frac{d}{dx} \sec x &= \sec x \tan x & \frac{d}{dx} \csc x &= -\csc x \cot x \end{aligned}$$

Theorem 1.1.6: Chain Rule

If g is differentiable at x and f is differentiable at $g(x)$, let $y = f(u)$ and $u = g(x)$, then

$$\frac{d}{dx}f(g(x)) = f'(g(x))g'(x) \quad \frac{dy}{dx} = \frac{dy}{du} \frac{du}{dx}$$

Proof

Assume first that $g(x) \neq g(a)$ for x sufficiently close to a , $x \neq a$. Then

$$(f \circ g)'(a) = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{x - a} = \lim_{x \rightarrow a} \frac{f(g(x)) - f(g(a))}{g(x) - g(a)} \cdot \frac{g(x) - g(a)}{x - a} = f'(g(a))g'(a)$$

□

Inverse Function Derivative inversefunctionderivative

$$(f^{-1})'(x) = \frac{1}{f'(f^{-1}(x))}$$

Exponential and Logarithm Derivatives

$$\begin{aligned} \frac{d}{dx} e^x &= e^x & \frac{d}{dx} a^x &= a^x \ln a \\ \frac{d}{dx} \ln x &= \frac{1}{x} & \frac{d}{dx} \log_a x &= \frac{1}{x \ln a} \end{aligned}$$

Inverse Trigonometric Derivatives

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}} \quad \frac{d}{dx} \arccos x = -\frac{1}{\sqrt{1-x^2}} \quad \frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

Implicit Derivative A function may be defined implicitly by a relation between x and y ; instead of solving for y explicitly, differentiate both sides with respect to x , treating y as a function of x , and apply the chain rule when differentiating expressions involving y , e.g. $\frac{d}{dx}[y] = y'$, $\frac{d}{dx}[y^2] = 2yy'$, $\frac{d}{dx}[\sin y] = \cos(y)y'$.

Linearization If f is differentiable at $x = a$, the **linearization** of f at a is $L(x) = f(a) + f'(a)(x - a)$.

Differential If $y = f(x)$, then $dy = f'(x) dx$. The actual change is $\Delta y = f(x + \Delta x) - f(x) \approx dy = f'(x)\Delta x$, $\Delta x \rightarrow 0$.

Theorem 1.1.7: Extreme Value Theorem

If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum and minimum on $[a, b]$.

Local and Absolute Extremum

- f has a **local maximum** at c if $f(c) \geq f(x)$ for all x sufficiently close to c .
- f has a **local minimum** at c if $f(c) \leq f(x)$ for all x sufficiently close to c .
- f has an **absolute maximum** at c on a domain D if $f(c) \geq f(x)$ for all $x \in D$.
- f has an **absolute minimum** at c on a domain D if $f(c) \leq f(x)$ for all $x \in D$.

Critical Number A number c in the domain of f is a **critical number** if $f'(c) = 0$ or $f'(c)$ does not exist.

Closed Interval Method To find absolute extrema of a continuous function f on $[a, b]$:

1. Find all critical numbers in (a, b) .
2. Evaluate f at the critical numbers. Evaluate $f(a)$ and $f(b)$. Compare all values.

Theorem 1.1.8: Rolle's Theorem

Suppose f is continuous on $[a, b]$, f is differentiable on (a, b) , and $f(a) = f(b)$, then $\exists c \in (a, b)$ such that

$$f'(c) = 0$$

Proof

Since f is continuous on $[a, b]$, by the Extreme Value Theorem, f attains a maximum and a minimum on $[a, b]$. If f is constant on $[a, b]$, then $f'(x) = 0$ for every $x \in (a, b)$, so the result holds. Assume f is not constant. Since $f(a) = f(b)$, at least one of the maximum or minimum values of f is attained at some interior point $c \in (a, b)$. Thus f has a local maximum or local minimum at c . Because f is differentiable on (a, b) , Fermat's Theorem gives $f'(c) = 0$. Therefore, there exists $c \in (a, b)$ such that $f'(c) = 0$. $\square$

Theorem 1.1.9: Mean Value Theorem

Suppose f is continuous on $[a, b]$ and f is differentiable on (a, b) then there exists $c \in (a, b)$ such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Proof

Let $m = \frac{f(b) - f(a)}{b - a}$. Consider the secant line $L(x) = f(a) + m(x - a)$, and define $g(x) = f(x) - L(x)$. Since f is continuous on $[a, b]$ and differentiable on (a, b) , and L is linear, g is also continuous on $[a, b]$ and differentiable on (a, b) . Moreover, $g(a) = f(a) - L(a) = f(a) - f(a) = 0$, and $g(b) = f(b) - L(b) = f(b) - (f(a) + m(b - a)) =$

$f(b) - (f(a) + f(b) - f(a)) = 0$. Thus $g(a) = g(b)$. By Rolle's Theorem, there exists $c \in (a, b)$ such that $g'(c) = 0$. But $g'(x) = f'(x) - L'(x) = f'(x) - m$. Therefore, $f'(c) = m = \frac{f(b)-f(a)}{b-a}$. $\square$

First Derivative Test Let c be a critical number:

1. If f' changes from positive to negative at c , then f has a local maximum at c .
2. If f' changes from negative to positive at c , then f has a local minimum at c .
3. If f' does not change sign at c , then f has no local extremum at c .

Concavity

1. f is **concave up** on an interval if $f''(x) > 0$ on that interval.
2. f is **concave down** on an interval if $f''(x) < 0$ on that interval.

Second Derivative Test Suppose $f'(c) = 0$:

1. If $f''(c) > 0$, then f has a local minimum at c .
2. If $f''(c) < 0$, then f has a local maximum at c .
3. If $f''(c) = 0$, the test is inconclusive.

Inflection Point A point $x = c$ is an **inflection point** if f is continuous at c and concavity changes at c . Possible candidates occur when $f''(c) = 0$ or $f''(c)$ does not exist

Theorem 1.1.10: L'Hopital's Rule

If $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$ or both limits are infinite, and if $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$ exists or equals $\pm\infty$, then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

1.2 Integrations and the Fundamental Theorem of Calculus

Area Approximation On $[a, b]$, $\Delta x = \frac{b-a}{n}$. Right endpoint: $x_i = a + i\Delta x$. Left endpoint: $x_{i-1} = a + (i-1)\Delta x$. Midpoint: $\bar{x}_i = a + (i - \frac{1}{2})\Delta x$. The approximate area is $\sum_{i=1}^n f(x_i^*)\Delta x$.

Definition 1.2.1: Definite Integral

If the limit exists, let $\Delta x = \frac{b-a}{n}$:

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*)\Delta x$$

A definite integral is signed area. Area above the x -axis is positive; area below is negative.

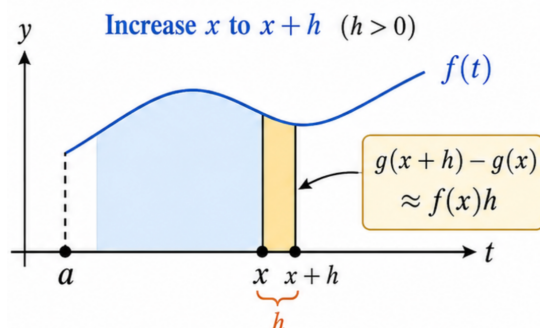
Properties

$$\begin{aligned} \int_a^a f(x) dx &= 0 & \int_a^b f(x) dx &= -\int_b^a f(x) dx & \int_a^b cf(x) dx &= c \int_a^b f(x) dx \\ \int_a^b [f(x) + g(x)] dx &= \int_a^b f(x) dx + \int_a^b g(x) dx & \int_a^b f(x) dx &= \int_a^c f(x) dx + \int_c^b f(x) dx \end{aligned}$$

Antiderivative F is an antiderivative of f if $F'(x) = f(x)$. All antiderivatives are $F(x) + C$.

Theorem 1.2.2: Fundamental Theorem of Calculus Part 1

If f is continuous and $g(x) = \int_a^x f(t) dt$ then $g'(x) = f(x)$.



The rate of change of accumulated area is the current height of the function.

Proof

By the definition of derivative, $g'(x) = \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h}$. Since $g(x+h) - g(x) = \int_a^{x+h} f(t) dt - \int_a^x f(t) dt = \int_x^{x+h} f(t) dt$, we have $g'(x) = \lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t) dt$. Because f is continuous, the average value of f on $[x, x+h]$ approaches $f(x)$ as $h \rightarrow 0$. Hence $\lim_{h \rightarrow 0} \frac{1}{h} \int_x^{x+h} f(t) dt = f(x)$. Therefore, $g'(x) = f(x)$. $\square$

Theorem 1.2.3: Fundamental Theorem of Calculus Part 2

If f is continuous on $[a, b]$ and $F'(x) = f(x)$, then

$$\int_a^b f(x) dx = F(b) - F(a)$$

Adding up infinitely many tiny changes $f(x)dx$ gives the total change $F(b) - F(a)$ from a to b .

Proof

Define $g(x) = \int_a^x f(t) dt$. By the Fundamental Theorem of Calculus Part 1, $g'(x) = f(x)$. Since $F'(x) = f(x)$, we have $g'(x) = F'(x)$. Therefore, for some constant C , $g(x) = F(x) + C$. Evaluating at $x = a$, we get $g(a) = \int_a^a f(t) dt = 0$. Hence $C = -F(a)$. Therefore, $g(x) = F(x) - F(a)$. Setting $x = b$, we obtain the result. $\square$

Substitution Rule If $u = g(x)$, $du = g'(x) dx$, then

$$\int f(g(x))g'(x) dx = \int f(u) du$$

For definite integrals, change the bounds:

$$\int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

Integration by Parts

$$\int u dv = uv - \int v du$$

For definite integrals,

$$\int_a^b u dv = [uv]_a^b - \int_a^b v du$$

Partial Fractions For a proper rational function $P(x)/Q(x)$ (where $\deg P < \deg Q$):

- **Distinct linear:** $(x-a)(x-b) \rightarrow \frac{A}{x-a} + \frac{B}{x-b}$
- **Repeated linear:** $(x-a)^r \rightarrow \frac{A_1}{x-a} + \cdots + \frac{A_r}{(x-a)^r}$
- **Irreducible quadratic:** $ax^2 + bx + c \rightarrow \frac{Ax+B}{ax^2+bx+c}$

Key integrals:

$$\int \frac{dx}{ax+b} = \frac{1}{a} \ln |ax+b| + C, \quad \int \frac{dx}{x^2+a^2} = \frac{1}{a} \arctan\left(\frac{x}{a}\right) + C$$

Infinite Intervals

$$\int_a^\infty f(x) dx = \lim_{t \rightarrow \infty} \int_a^t f(x) dx$$

$$\int_{-\infty}^b f(x) dx = \lim_{t \rightarrow -\infty} \int_t^b f(x) dx$$

For $\int_{-\infty}^\infty f(x) dx$ split at any c :

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^c f(x) dx + \int_c^\infty f(x) dx.$$

Comparison Theorem If $0 \leq f(x) \leq g(x)$ on $[a, \infty)$, then:

1. If $\int_a^\infty g(x) dx$ converges, then $\int_a^\infty f(x) dx$ converges.
2. If $\int_a^\infty f(x) dx$ diverges, then $\int_a^\infty g(x) dx$ diverges.

Absolute Area If f crosses the x -axis, geometric area is below, split at zeros of f and remove negative signed area:

$$A = \int_a^b |f(x)| dx$$

Area Between Curves If $f(x) \geq g(x)$ on $[a, b]$, then

$$A = \int_a^b [f(x) - g(x)] dx$$

Surface Area If $y = f(x) \geq 0$ and f' is continuous on $[a, b]$, then rotating the curve around the x -axis gives

$$S = 2\pi \int_a^b f(x) \sqrt{1 + [f'(x)]^2} dx$$

1.3 Sequences, Series, and Power Series

Limit of a sequence A sequence (a_n) converges to $L \in \mathbb{R}$ if for every $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that

$$n \geq N \implies |a_n - L| < \varepsilon \implies \lim_{n \rightarrow \infty} a_n = L$$

Convergence of a series The series $\sum_{n=1}^{\infty} a_n$ converges to S if $\lim_{N \rightarrow \infty} s_N = S$.

- Geometric series, converges if $|r| < 1$: $\sum_{n=0}^{\infty} ar^n = \frac{a}{1-r}$.
- Harmonic series, diverges: $\sum_{n=1}^{\infty} \frac{1}{n}$.

A necessary condition for convergence is $\sum a_n$ converges $\implies \lim_{n \rightarrow \infty} a_n = 0$. The converse is false.

Integral Test Let f be continuous, positive, and decreasing on $[1, \infty)$, and let $a_n = f(n)$. Then $\sum_{n=1}^{\infty} a_n$ converges if and only if $\int_1^{\infty} f(x) dx$ converges.

Direct Comparison Test Suppose $0 \leq a_n \leq b_n$ for all sufficiently large n .

- If $\sum b_n$ converges, then $\sum a_n$ converges.
- If $\sum a_n$ diverges, then $\sum b_n$ diverges.

Limit Comparison Test Let $a_n, b_n > 0$ and suppose $\lim_{n \rightarrow \infty} \frac{a_n}{b_n} = c$ where $0 < c < \infty$, $\sum a_n, \sum b_n$ either both converge or both diverge.

Alternating Series An alternating series has the form

$$\sum_{n=1}^{\infty} (-1)^{n-1} b_n, \quad b_n \geq 0$$

Alternating Series Test If $b_{n+1} \leq b_n$ for all sufficiently large n and $\lim_{n \rightarrow \infty} b_n = 0$, then $\sum_{n=1}^{\infty} (-1)^{n-1} b_n$ converges.

Absolute and conditional convergence The series $\sum a_n$ converges absolutely if $\sum |a_n|$ converges. It converges conditionally if $\sum a_n$ converges but $\sum |a_n|$ diverges. Absolute convergence is stronger than ordinary convergence:

$$\sum |a_n| \text{ converges} \implies \sum a_n \text{ converges}$$

Ratio test For a series $\sum a_n$, suppose $L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$, then:

$$L < 1 \Rightarrow \text{absolute convergence}, \quad L > 1 \Rightarrow \text{divergence}, \quad L = 1 \Rightarrow \text{inconclusive}$$

Root test For a series $\sum a_n$, suppose $L = \lim_{n \rightarrow \infty} \sqrt[n]{|a_n|}$, then

$$L < 1 \Rightarrow \text{absolute convergence}, \quad L > 1 \Rightarrow \text{divergence}, \quad L = 1 \Rightarrow \text{inconclusive}$$

Power Series A power series centered at a has the form

$$\sum_{n=0}^{\infty} c_n (x - a)^n$$

There exists a radius of convergence $R \in [0, \infty]$ such that the series converges for $|x - a| < R$ and diverges for $|x - a| > R$. Endpoints must be checked separately. The interval of convergence is obtained by:

1. applying the ratio or root test to find R
2. checking $x = a - R$ and $x = a + R$ individually

Definition 1.3.1: Taylor and Maclaurin Series

If f is infinitely differentiable near a , its Taylor series at a is below. $a = 0$ gives a Maclaurin series.

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

Proof

The idea of Taylor expansion is to approximate $f(x)$ near a by a polynomial $P_k(x) = c_0 + c_1(x - a) + c_2(x - a)^2 + \cdots + c_k(x - a)^k$. We choose the coefficients so that P_k has the same derivatives as f at a : $P_k^{(j)}(a) = f^{(j)}(a)$, $j = 0, 1, \dots, k$. Since $P_k^{(j)}(a) = j!c_j$, we get $c_j = \frac{f^{(j)}(a)}{j!}$. Therefore the k -th Taylor polynomial of f at a is $P_k(x) = \sum_{j=0}^k \frac{f^{(j)}(a)}{j!} (x - a)^j$. $\square$

The most important standard Maclaurin series are:

$$\begin{aligned} e^x &= \sum_{n=0}^{\infty} \frac{x^n}{n!}, & \sin x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} \\ \cos x &= \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!}, & \frac{1}{1-x} &= \sum_{n=0}^{\infty} x^n, \quad |x| < 1 \end{aligned}$$

Theorem 1.3.2: Taylor Remainder Theorem

Suppose f is $k+1$ times differentiable on an interval containing a and x , there exists some δ between a and x such that

$$f(x) = \sum_{j=0}^k \frac{f^{(j)}(a)}{j!} (x - a)^j + \frac{f^{(k+1)}(\delta)}{(k+1)!} (x - a)^{k+1}$$

Proof

Assume $a < x$. The case $x < a$ is similar. Define R by

$$f(x) = \sum_{j=0}^k \frac{f^{(j)}(a)}{j!} (x - a)^j + \frac{R}{(k+1)!} (x - a)^{k+1}$$

We need to show that $R = f^{(k+1)}(\delta)$ for some $\delta \in (a, x)$. Define, for $z \in [a, x]$,

$$g(z) = \sum_{j=0}^k \frac{f^{(j)}(z)}{j!} (x - z)^j + \frac{R}{(k+1)!} (x - z)^{k+1}$$

Then $g(a) = f(x), g(x) = f(x)$. Hence by Rolle's theorem, there exists $\delta \in (a, x)$ such that $g'(\delta) = 0$.

$$\begin{aligned} g'(z) &= \sum_{j=0}^k \frac{f^{(j+1)}(z)}{j!} (x-z)^j - \sum_{j=1}^k \frac{f^{(j)}(z)}{(j-1)!} (x-z)^{j-1} - \frac{R}{k!} (x-z)^k \\ &= \frac{f^{(k+1)}(z)}{k!} (x-z)^k - \frac{R}{k!} (x-z)^k \\ &= \frac{(x-z)^k}{k!} (f^{(k+1)}(z) - R) \end{aligned}$$

Since $g'(\delta) = 0$, we have $\frac{(x-\delta)^k}{k!} (f^{(k+1)}(\delta) - R) = 0$. Because $\delta \in (a, x)$, $x - \delta \neq 0$. Therefore $R = f^{(k+1)}(\delta)$. $\square$

1.4 Functions of Several Variables

3D Coordinate Systems A point in three-dimensional space is written as (x, y, z) . The distance between two points $P_1 = (x_1, y_1, z_1)$ and $P_2 = (x_2, y_2, z_2)$ is:

$$d(P_1, P_2) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}.$$

Vectors A vector is a quantity with magnitude and direction:

$$\mathbf{v} = \langle v_1, v_2, v_3 \rangle \quad \|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2 + v_3^2}$$

Dot Product The dot product of $\mathbf{a} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{b} = \langle b_1, b_2, b_3 \rangle$ is

$$\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3 = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

Projections The projection of $\mathbf{a}$ onto $\mathbf{b}$ is

$$\text{proj}_{\mathbf{b}} \mathbf{a} = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{b}\|^2} \mathbf{b}$$

Cross Product The cross product is below which is perpendicular to both $\mathbf{a}$ and $\mathbf{b}$

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

$$\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta$$

Line A line through a point $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$ in direction $\mathbf{v} = \langle a, b, c \rangle$ is

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}$$

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

Plane A plane through (x_0, y_0, z_0) with normal vector $\mathbf{n} = \langle A, B, C \rangle$ is

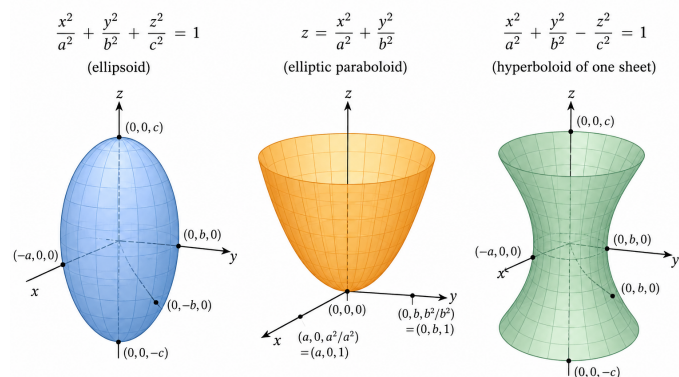
$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Quadric Surface Quadric surfaces are second-degree surfaces in x, y, z . Common models include:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1 \quad (\text{ellipsoid})$$

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2} \quad (\text{elliptic paraboloid})$$

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1 \quad (\text{hyperboloid of one sheet})$$



Vector Functions A vector-valued function has the following form, which describes a curve in space.

$$\mathbf{r}(t) = \langle f(t), g(t), h(t) \rangle$$

The derivative is componentwise, which is tangent to the curve.

$$\mathbf{r}'(t) = \langle f'(t), g'(t), h'(t) \rangle$$

$$\frac{d}{dt}(\mathbf{u} \cdot \mathbf{v}) = \mathbf{u}' \cdot \mathbf{v} + \mathbf{u} \cdot \mathbf{v}' \quad \frac{d}{dt}(\mathbf{u} \times \mathbf{v}) = \mathbf{u}' \times \mathbf{v} + \mathbf{u} \times \mathbf{v}'$$

Arc Length The length of a smooth curve $\mathbf{r}(t)$ for $a \leq t \leq b$ is

$$L = \int_a^b \|\mathbf{r}'(t)\| dt$$

Curvature Curvature measures how fast the tangent direction changes with respect to arc length:

$$\kappa = \left\| \frac{d\mathbf{T}}{ds} \right\| \quad \mathbf{T}(t) = \frac{\mathbf{r}'(t)}{\|\mathbf{r}'(t)\|}$$

A practical formula is

$$\kappa(t) = \frac{\|\mathbf{r}'(t) \times \mathbf{r}''(t)\|}{\|\mathbf{r}'(t)\|^3}$$

For a plane curve $y = f(x)$,

$$\kappa(x) = \frac{|f''(x)|}{(1 + [f'(x)]^2)^{3/2}}$$

Functions of Several Variables A function maps points in a domain to real numbers: $f(x, y)$ or $f(x, y, z)$. Its graph is the surface $z = f(x, y)$. A level curve $f(x, y) = c$ shows where the function has a constant value. A limit $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = L$ exists if f approaches L along every path to (a, b) . A function is continuous at (a, b) if $\lim_{(x,y) \rightarrow (a,b)} f(x, y) = f(a, b)$.

Partial Derivatives These measure rates of change in the x or y direction:

$$f_x(x, y) = \lim_{h \rightarrow 0} \frac{f(x+h, y) - f(x, y)}{h} \quad f_y(x, y) = \lim_{h \rightarrow 0} \frac{f(x, y+h) - f(x, y)}{h}$$

Linear Approximation For $z = f(x, y)$, the tangent plane is the linear approximation:

$$z = f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

$$f(x, y) \approx f(a, b) + f_x(a, b)(x - a) + f_y(a, b)(y - b)$$

Chain Rule For a composition $z = f(x(t), y(t))$, the chain rule is

$$\frac{dz}{dt} = f_x \frac{dx}{dt} + f_y \frac{dy}{dt}$$

For $z = f(x(s, t), y(s, t))$,

$$\frac{\partial z}{\partial s} = f_x \frac{\partial x}{\partial s} + f_y \frac{\partial y}{\partial s} \quad \frac{\partial z}{\partial t} = f_x \frac{\partial x}{\partial t} + f_y \frac{\partial y}{\partial t}$$

Definition 1.4.1: Directional Derivatives

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$, $\mathbf{a} \in \mathbb{R}^n$, $\mathbf{u} \in \mathbb{R}^n$, $\|\mathbf{u}\| = 1$. The directional derivative of f at $\mathbf{a}$ in the direction $\mathbf{u}$ is defined by

$$D_{\mathbf{u}}f(\mathbf{a}) = \lim_{h \rightarrow 0} \frac{f(\mathbf{a} + h\mathbf{u}) - f(\mathbf{a})}{h}$$

Definition 1.4.2: Gradient

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be differentiable at a point $\mathbf{a} = (a_1, \dots, a_n) \in \mathbb{R}^n$. The gradient of f at $\mathbf{a}$ is:

$$\nabla f(\mathbf{a}) = \left\langle \frac{\partial f}{\partial x_1}(\mathbf{a}), \frac{\partial f}{\partial x_2}(\mathbf{a}), \dots, \frac{\partial f}{\partial x_n}(\mathbf{a}) \right\rangle$$

The gradient of f at $\mathbf{a}$ is the vector of partial derivatives, and for any unit vector $\mathbf{u}$ the directional derivative in the direction of $\mathbf{u}$ is given by the dot product of the gradient with $\mathbf{u}$:

$$D_{\mathbf{u}}f(a, b) = \nabla f(a, b) \cdot \mathbf{u}$$

Proof

Let $\mathbf{u} = \langle u_1, u_2 \rangle$ be a unit vector. Since f is differentiable at (a, b) , for $(\Delta x, \Delta y) \rightarrow (0, 0)$,

$$f(a + \Delta x, b + \Delta y) = f(a, b) + f_x(a, b)\Delta x + f_y(a, b)\Delta y + o\left(\sqrt{(\Delta x)^2 + (\Delta y)^2}\right)$$

Take $\Delta x = hu_1$ and $\Delta y = hu_2$. Since $\|\mathbf{u}\| = 1$,

$$f(a + hu_1, b + hu_2) = f(a, b) + hf_x(a, b)u_1 + hf_y(a, b)u_2 + o(|h|)$$

Therefore,

$$\begin{aligned} D_{\mathbf{u}}f(a, b) &= \lim_{h \rightarrow 0} \frac{f(a + hu_1, b + hu_2) - f(a, b)}{h} \\ &= \lim_{h \rightarrow 0} \left(f_x(a, b)u_1 + f_y(a, b)u_2 + \frac{o(|h|)}{h} \right) \\ &= f_x(a, b)u_1 + f_y(a, b)u_2 \\ &= \nabla f(a, b) \cdot \mathbf{u} \end{aligned}$$

□

The gradient points in the direction of **steepest ascent**, and the maximum possible directional derivative is $\|\nabla f(\mathbf{a})\|$. Thus, $\nabla f(\mathbf{a})$ gives both the direction and the rate of fastest increase of f at $\mathbf{a}$.

Proof

Assume f is differentiable at $\mathbf{a}$. Then for any $\|\mathbf{u}\| = 1$, $D_{\mathbf{u}}f(\mathbf{a}) = \nabla f(\mathbf{a}) \cdot \mathbf{u}$. By the Cauchy–Schwarz inequality, $\nabla f(\mathbf{a}) \cdot \mathbf{u} \leq \|\nabla f(\mathbf{a})\| \|\mathbf{u}\| = \|\nabla f(\mathbf{a})\|$. Equality holds when $\mathbf{u} = \frac{\nabla f(\mathbf{a})}{\|\nabla f(\mathbf{a})\|}$, provided $\nabla f(\mathbf{a}) \neq \mathbf{0}$. $\square$

Definition 1.4.3: Jacobian Matrix

Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be a function. If all first-order partial derivatives of $F_1, \dots, F_m$ exist at $\mathbf{a} \in \mathbb{R}^n$, then the Jacobian matrix of F at $\mathbf{a}$ is the $m \times n$ matrix

$$J_F(\mathbf{a}) = \begin{bmatrix} \frac{\partial F_1}{\partial x_1}(\mathbf{a}) & \frac{\partial F_1}{\partial x_2}(\mathbf{a}) & \cdots & \frac{\partial F_1}{\partial x_n}(\mathbf{a}) \\ \frac{\partial F_2}{\partial x_1}(\mathbf{a}) & \frac{\partial F_2}{\partial x_2}(\mathbf{a}) & \cdots & \frac{\partial F_2}{\partial x_n}(\mathbf{a}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_m}{\partial x_1}(\mathbf{a}) & \frac{\partial F_m}{\partial x_2}(\mathbf{a}) & \cdots & \frac{\partial F_m}{\partial x_n}(\mathbf{a}) \end{bmatrix} = \left[\frac{\partial F_i}{\partial x_j}(\mathbf{a}) \right]_{1 \leq i \leq m, 1 \leq j \leq n} \quad F(\mathbf{x}) = \begin{bmatrix} F_1(\mathbf{x}) \\ F_2(\mathbf{x}) \\ \vdots \\ F_m(\mathbf{x}) \end{bmatrix} \quad \mathbf{x} = (x_1, \dots, x_n)$$

Definition 1.4.4: Hessian Matrix

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a scalar-valued function, where $\mathbf{x} = (x_1, \dots, x_n)$. If all second-order partial derivatives of f exist at $\mathbf{a} \in \mathbb{R}^n$, then the Hessian matrix of f at $\mathbf{a}$ is the $n \times n$ matrix

$$H_f(\mathbf{a}) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2}(\mathbf{a}) & \frac{\partial^2 f}{\partial x_1 \partial x_2}(\mathbf{a}) & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n}(\mathbf{a}) \\ \frac{\partial^2 f}{\partial x_2 \partial x_1}(\mathbf{a}) & \frac{\partial^2 f}{\partial x_2^2}(\mathbf{a}) & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n}(\mathbf{a}) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1}(\mathbf{a}) & \frac{\partial^2 f}{\partial x_n \partial x_2}(\mathbf{a}) & \cdots & \frac{\partial^2 f}{\partial x_n^2}(\mathbf{a}) \end{bmatrix} = \left[\frac{\partial^2 f}{\partial x_i \partial x_j}(\mathbf{a}) \right]_{1 \leq i, j \leq n}$$

Unconstrained Extrema Critical points satisfy $\nabla f = \mathbf{0}$. For a function of two variables, define

$$D = f_{xx}f_{yy} - (f_{xy})^2 = |H_f|$$

Then the second derivative test says:

- if $D > 0$ and $f_{xx} > 0$, there is a local minimum;
- if $D > 0$ and $f_{xx} < 0$, there is a local maximum;
- if $D < 0$, there is a saddle point;
- if $D = 0$, the test is inconclusive.

Lagrange Multipliers To optimize $f(x, y)$ subject to $g(x, y) = c$, solve

$$\nabla f = \lambda \nabla g, \quad g(x, y) = c$$

For two constraints $g(x, y, z) = c$ and $h(x, y, z) = d$, solve

$$\nabla f = \lambda \nabla g + \mu \nabla h, \quad g = c, \quad h = d$$

Evaluate f at all candidate points, especially when the constraint set is closed and bounded.

1.5 Multiple Integrals and Coordinate Systems

Double Integrals For a function $f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$, the double integral is

$$\iint_R f(x, y) \, dA$$

It gives the signed volume under $z = f(x, y)$ over R . Fubini's theorem allows computation by iterated integrals:

$$\iint_R f(x, y) \, dA = \int_a^b \int_c^d f(x, y) \, dy \, dx = \int_c^d \int_a^b f(x, y) \, dx \, dy$$

For a general region D , one writes the bounds according to the shape of the domain. A Type I region has

$$D = \{(x, y) : a \leq x \leq b, g_1(x) \leq y \leq g_2(x)\} \quad \iint_D f \, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x, y) \, dy \, dx$$

A Type II region has

$$D = \{(x, y) : c \leq y \leq d, h_1(y) \leq x \leq h_2(y)\} \quad \iint_D f \, dA = \int_c^d \int_{h_1(y)}^{h_2(y)} f(x, y) \, dx \, dy$$

Polar coordinates are $x = r \cos \theta, y = r \sin \theta$, with area element $dA = r \, dr \, d\theta$. Thus

$$\iint_D f(x, y) \, dA = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} f(r \cos \theta, r \sin \theta) r \, dr \, d\theta$$

Triple Integrals A triple integral over a solid region E is

$$\iiint_E f(x, y, z) \, dV.$$

It can represent mass when f is density. Rectangular coordinates use $dV = dx \, dy \, dz$. Cylindrical coordinates are $x = r \cos \theta, y = r \sin \theta, z = z$ with $dV = r \, dz \, dr \, d\theta$. Spherical coordinates are $x = \rho \sin \phi \cos \theta, y = \rho \sin \phi \sin \theta, z = \rho \cos \phi$, with volume element $dV = \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta$.

Vector Field A vector field assigns a vector to each point:

$$\mathbf{F}(x, y) = P(x, y)\mathbf{i} + Q(x, y)\mathbf{j}$$

Line Integral For a scalar function f or a vector field $\mathbf{F}$ along a curve C parametrized by $\mathbf{r}(t)$, $a \leq t \leq b$,

$$\int_C f \, ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| \, dt$$

$$\int_C \mathbf{F} \cdot d\mathbf{r} = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) \, dt$$

The fundamental theorem for line integrals says that if $\mathbf{F} = \nabla f$ is conservative, then:

$$\int_C \mathbf{F} \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

Theorem 1.5.1: Green's theorem

Let C be a positively oriented, piecewise smooth, simple closed curve enclosing a region D . If P and Q have continuous partial derivatives on an open region containing D , then

$$\oint_C P dx + Q dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dA$$

Parametric Surface A parametric surface is described by

$$\mathbf{r}(u, v) = \langle x(u, v), y(u, v), z(u, v) \rangle, \quad (u, v) \in D$$

For fixed u or fixed v , one obtains parameter curves on the surface. The tangent vectors are $\mathbf{r}_u = \frac{\partial \mathbf{r}}{\partial u}$, $\mathbf{r}_v = \frac{\partial \mathbf{r}}{\partial v}$. A normal vector is $\mathbf{r}_u \times \mathbf{r}_v$. The tangent plane at (u_0, v_0) is determined by the point $\mathbf{r}(u_0, v_0)$ and the normal vector $\mathbf{r}_u(u_0, v_0) \times \mathbf{r}_v(u_0, v_0)$. The surface area is

$$A(S) = \iint_D \|\mathbf{r}_u \times \mathbf{r}_v\| dA$$

Surface Integral For a scalar function f over a parametric surface S given by $\mathbf{r}(u, v)$

$$\iint_S f dS = \iint_D f(\mathbf{r}(u, v)) \|\mathbf{r}_u \times \mathbf{r}_v\| dA$$

For a vector field $\mathbf{F}$, the flux across an oriented surface S is

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iint_D \mathbf{F}(\mathbf{r}(u, v)) \cdot (\mathbf{r}_u \times \mathbf{r}_v) dA$$

Theorem 1.5.2: Divergence theorem

Let E be a solid region with positively oriented closed boundary surface $S = \partial E$. Then

$$\iint_S \mathbf{F} \cdot d\mathbf{S} = \iiint_E \operatorname{div} \mathbf{F} dV.$$